GIS TERM PROJECT · FINAL PRESENTATION

Geospatial Reasoning using LLMs and Generative AI

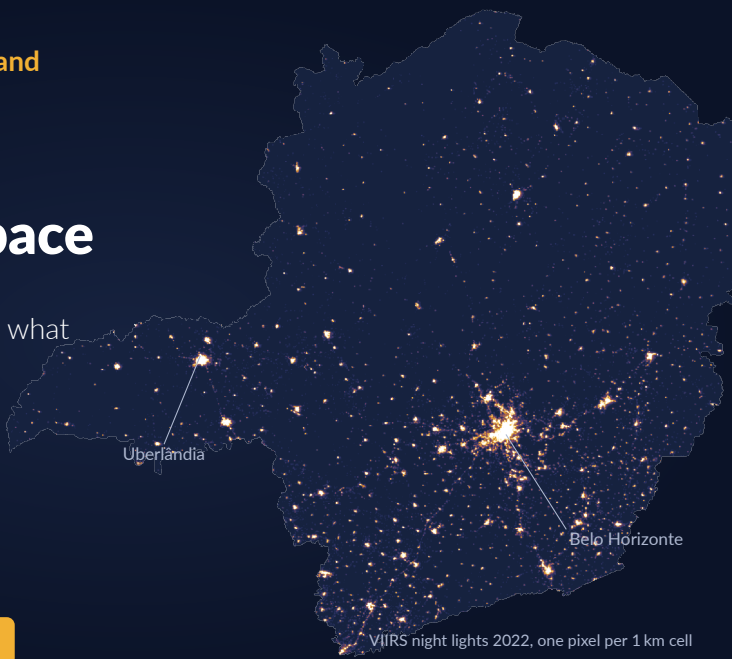
Reading the economy from space

Municipal GDP from satellite embeddings, text and night lights: what we learned since the mid-term

Group 2

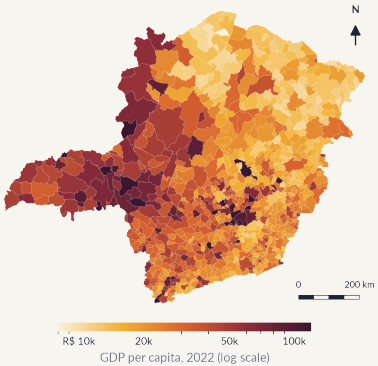
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github.com/chadsaras/GIS_Project



VIIRS night lights 2022, one pixel per 1 km cell

The question: can free satellite data estimate local GDP?



49× gap in GDP per capita inside one state

Catas Altas (R$ 391k) vs São João das Missões (R$ 8k)

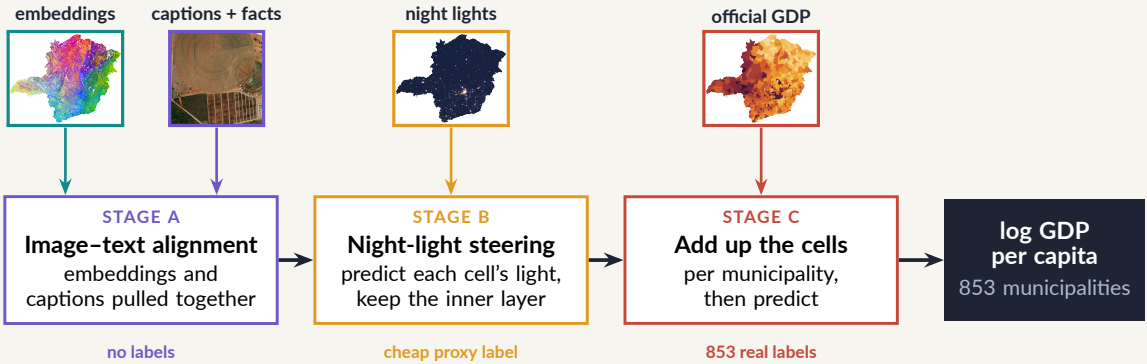
OBJECTIVE

Estimate 2022 GDP per capita for all **853 municipalities** of Minas Gerais from **free satellite data**, combining two recent ideas: ● **AI captions** (UrbanCLIP) and ● **night-light steering** of ● **satellite embeddings** (Yue et al., 2026), and test it **honestly**.

RESEARCH QUESTIONS

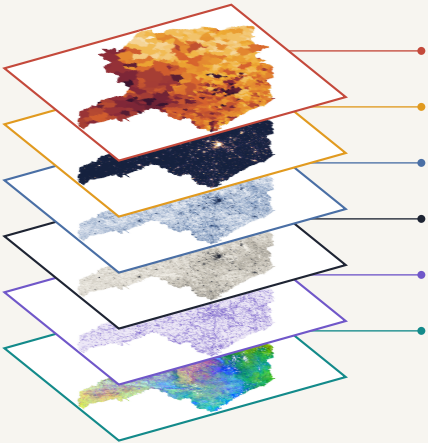
RQ1	Does night-light steering improve the estimate?
RQ2	Does aligning with text (captions + facts) help?
RQ3	Do points of interest and roads add information?
RQ4	Does a hybrid with night lights beat the best model?
RQ5	Where and why does the model fail?

Recap: three learning stages, from pixels to official GDP



Stage A: 64→128 projections, contrastive loss. Stage B: 128→64→**32**→1, the 32-number layer is the steered embedding. Stage C: per-cell network, log-sum pooling, 5-seed ensemble. Captions are needed only in training.

Recap: six free layers and 10,000 AI captions



GDP per capita (target)
IBGE 2022, official

Night lights
VIIRS 2022

Roads and POIs
OpenStreetMap

Buildings
Microsoft + OSM footprints

Captioned tiles
10,000 Esri tiles, Qwen2.5-VL

Satellite embeddings
Google GSED, 64 numbers, 10 m to 1 km

575k

kept 1 km cells, 853 municipalities

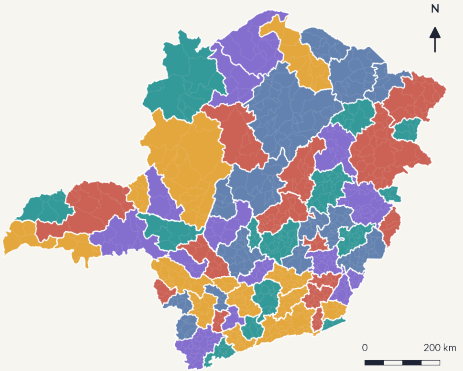
TEXT, PER SAMPLED CELL

- **Caption** by Qwen2.5-VL-7B (Kaggle, short prompt after a 200-tile pilot)
- **Fact sentences** from the data: land cover, POIs, roads (no night lights)
- **Cleaning**: filler and claims contradicted by map data removed; RemoteCLIP drops 246 captions

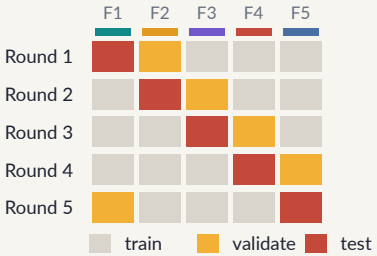
84%

of 60,023 caption sentences kept

Recap: every town is tested in a region never seen



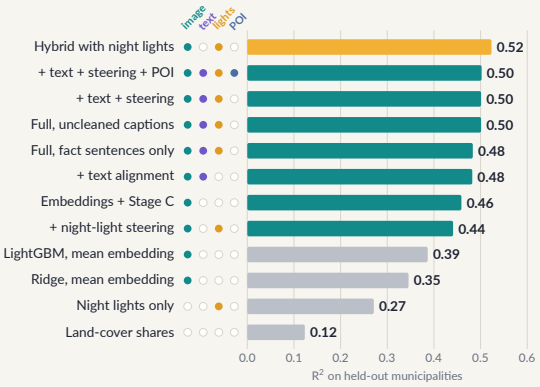
70 official regions (white outlines) grouped into 5 folds



- Whole regions held out: **no look-alike neighbour** in training. All three stages retrained each round.
- Scalers, tuning and blend weights use training and validation only: **nothing is tuned on test towns**.

Models are compared on the same folds with a block bootstrap over the 70 regions (95% intervals).

Recap: at the mid-term, satellites explained half of GDP



R^2 on held-out municipalities. Dots: ingredients each model uses. Hybrid = best model blended with night lights.

RQ STATUS AT THE MID-TERM

RQ1 STEERING no clear effect

RQ2 TEXT largest gain, borderline

RQ3 POI, ROADS nothing added

RQ4 HYBRID best, not significant

RQ5 FAILURES not answered

OPEN AT THE MID-TERM

Is the result **reproducible**? How good are the **captions**? **Where and why** does it fail? What did each part **learn**? Does it work **elsewhere**?

Lesson already then: a calibration fix (−21% error) mattered more than steering (backup B2).

Since the mid-term: seven questions, seven pieces of work

#	QUESTION	WORK DONE	ANSWER
1	Is the result reproducible?	Full rerun from a fresh clone and data folder	Same verdicts; text gain shrinks
2	How good are the AI captions?	Blind audit of 300 captions, 1,801 sentences	35% have a false claim, 23% after cleaning
3	Where does it fail? (RQ5)	Maps, ten largest misses, reasons	Mines, steel mills, commuter towns
4	Why does it fail? (RQ5)	Error regression, LOWESS, Moran's I, seeds	Errors rise with income, half as fast as night lights
5	Did each stage do its job?	Stage A/B diagnostics, neighbour check	Yes, but steering does not reach GDP
6	What did the model learn?	Text explanations; town-vector twins and clusters	Text reads geography; vectors carry size, not wealth
7	Does it work elsewhere?	São Paulo, models applied unchanged	Satellites R^2 0.33, night lights −0.06

Also since the mid-term: a 12-page report, results frozen (tag v1.0-results), a one-command rerun script and a data release.

A full rerun from scratch gives the same answers

NEW SINCE MID-TERM

RESULTS: FROZEN VS RERUN

Model	R^2 frozen	R^2 rerun	ΔRMSE
V1 night lights	0.271	0.271	0.000
V3 embeddings + C	0.459	0.469	−0.004
V4 + steering	0.441	0.489	−0.020
V6 + text + steering	0.501	0.509	−0.004
V7 full model	0.502	0.519	−0.007
V10 hybrid	**0.523**	**0.522**	0.000
V12 LightGBM	0.386	0.386	0.000

How: `run_all.sh` on a fresh clone and an empty data folder, every step from the raw downloads (about 3 days). Only the Earth Engine rasters, caption tiles and captions were reused.

RESEARCH QUESTIONS: ΔRMSE

Question	frozen	rerun	verdict
RQ1 steering	+0.007	−0.008	same
RQ2 text	**−0.025**	**−0.009**	same
RQ3 POI, roads	−0.001	−0.004	same
RQ4 hybrid	−0.009	−0.002	same

Best model unchanged (R^2 0.52). Every interval includes 0 in both runs.

The text gain is not robust: it shrinks from −0.025 to −0.009.

Deterministic steps match exactly; neural models differ by up to 0.02 RMSE (multi-threaded CPU training).

Blind audit: a third of AI captions hold a false claim

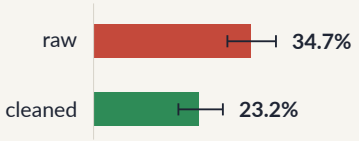
NEW SINCE MID-TERM

METHOD

1. 300 random captions, **1,801 sentences**
2. Each sentence labelled **correct**, **false** or **vague**, with the tile in view
3. Labeller **blind** to what cleaning removed
4. Raw and cleaned scores from the same labels; Wilson 95% intervals

Labels: 1,440 correct, 235 vague, 126 false. Labelled by an LLM (Claude); a human check is still open.

CAPTIONS WITH ≥1 FALSE CLAIM



Only 7% of sentences are false, but a caption has about six, so a third of captions contain one.

HOW WELL CLEANING WORKS

41%

of false sentences removed (recall)

58%

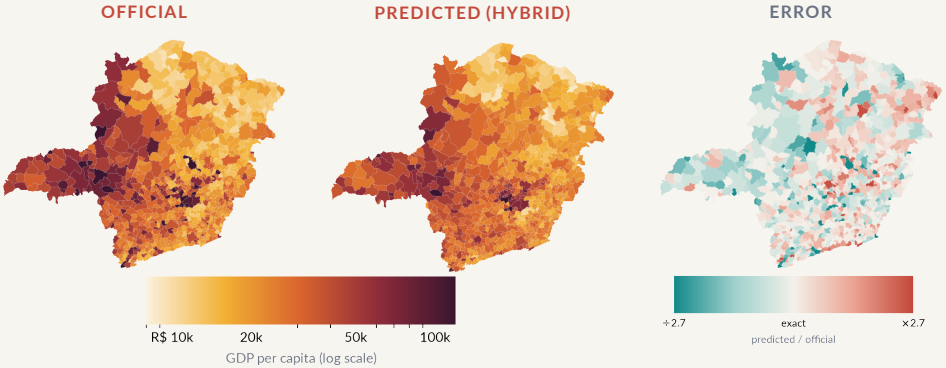
of removed sentences were false or vague (precision)

TYPICAL FALSE CLAIMS

“no buildings” next to farmhouses; “farmland” on scrub; “mining” on centre-pivot fields; ponds that are not there

RQ5: predictions follow the map, but are pulled to the middle

NEW SINCE MID-TERM



75%

of municipalities predicted within a factor 1.5 of official

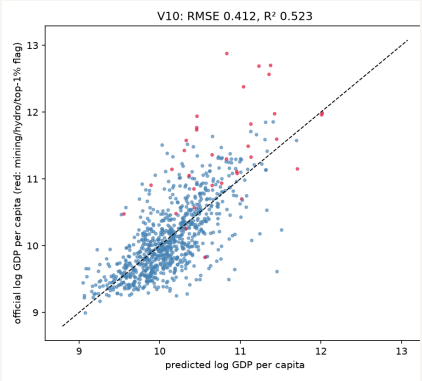
Under-predicted: rich agribusiness and mining towns (west, centre).

Over-predicted: small public-sector and commuter towns.

Every value is out-of-sample: each town predicted in the round where its region was held out.

RQ5: the biggest misses are mines and steel mills

NEW SINCE MID-TERM



Hybrid, 853 test predictions; red = flagged (industry over 60% of the economy, or top 1%).

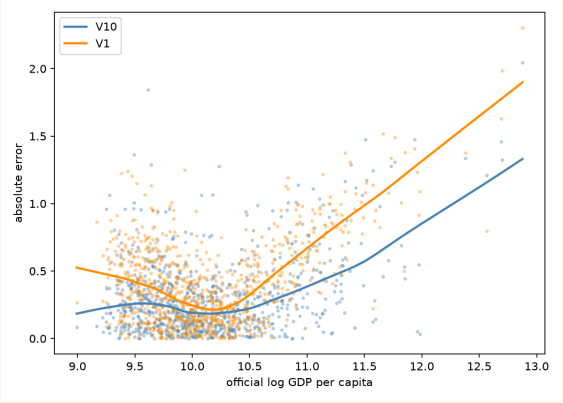
TEN LARGEST GAPS

Municipality	gap	likely reason
Catas Altas	-2.04	iron-ore mine
Santa Cruz de Minas	$+1.84$	tiny commuter town
Ressaquinha	-1.48	large plant
Conceição do Mato Dentro	-1.48	iron-ore mine
S. Gonçalo do Rio Abaixo	-1.46	iron-ore mine
Mário Campos	$+1.37$	BH commuter town
Itatiaiuçu	-1.34	iron-ore mine
Extrema	-1.32	logistics hub
Ouro Branco	-1.31	steel mill
Chiador	$+1.29$	public-sector town

Gap = predicted − official (log): -2.04 means 7.7× too low. A few cells make most of the output, while the tiles show forest and pasture. Reasons are candidates from sector shares.

RQ5: errors grow with income, half as fast as night lights

NEW SINCE MID-TERM



Absolute error against official income, LOWESS. Blue: hybrid; orange: night lights only.

0.08 vs **0.19**

extra error per log point of income, hybrid vs night lights

0.16 vs **0.33**

Moran's I of errors: still clustered in space, half as much

$\rho = 0.04$

spread across the 5 training seeds does not predict the error

Error regression with region-clustered SEs (backup B3): errors rise with income and built-up share, fall with services share; OSM completeness does not matter.

Each stage did its job, yet steering does not reach GDP

NEW SINCE MID-TERM

STAGE A: IMAGE–TEXT ALIGNMENT

Text used	recall@10	built-up probe R^2
cleaned captions + facts	0.52	0.67
raw captions	0.35	0.68
facts only	0.55	0.64
chance	*0.04*	

STAGE B: NIGHT-LIGHT PREDICTION

validation R^2 **0.75–0.78** for every input (best: raw embeddings)

Mean over 5 rounds, validation towns. Recall within groups of 256 pairs. 81% of cells are unlit.

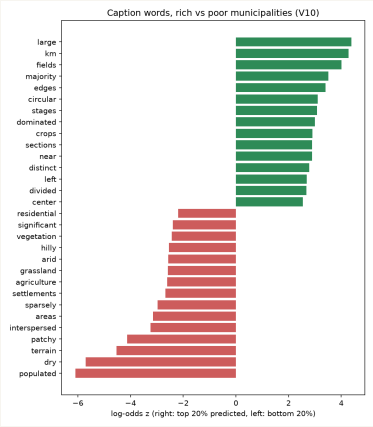
THE EMBEDDING PAPER'S NEIGHBOUR CHECK (ROUND 0)

	distance to 100 nearest		light gap to them	
Reference cell	raw	steered	raw	steered
Rural pasture	99 km	269 km	0.00	0.03
Small town	82 km	292 km	0.66	**0.46**
Industrial area	19 km	39 km	0.22	0.22
Mining / bare ground	49 km	96 km	0.90	**0.50**
BH suburb	8 km	28 km	0.20	0.20

Steered neighbours are **farther away but as active or more**: steering does what the paper says. Yet it adds nothing to GDP (RQ1).

The text space speaks the state's economic geography

NEW SINCE MID-TERM



RICHEST 20% PREDICTED

large, *fields*, *circular*, *crops*, *divided*: mechanised farming and centre pivots of the Triângulo

POOREST 20% PREDICTED

sparsely populated, *dry*, *terrain*, *patchy*, *arid*: the semi-arid north and Jequitinhonha

NEAREST TRAINING CAPTIONS TO A TOWN

Belo Horizonte (official 10.82, predicted 10.91): dense grids of red-tiled roofs.
São João das Missões, the poorest (8.99 vs 9.08): sparse settlements in dry scrub.
Conceição do Mato Dentro, a mine (11.94 vs 10.46): forest and pasture, so the mine stays invisible.

Word contrast: log-odds with informative Dirichlet prior (Monroe et al., 2008). Values: log GDP per capita.

Learned town vectors carry economy size, not wealth

NEW SINCE MID-TERM

TWINS: 10 MOST SIMILAR TRAINING TOWNS

Space	R^2 income/person	R^2 population
Raw satellite vectors	0.31	0.06
Population only	0.00	0.99
Stage C, as trained	0.22–0.33	0.74–0.81
Stage C, size removed	0.28–0.41	0.19–0.47
The GDP model itself	*0.46–0.52*	

Twin estimate = mean official value of a held-out town's 10 nearest training towns.

CLUSTERS: K-MEANS, HELD-OUT TOWNS

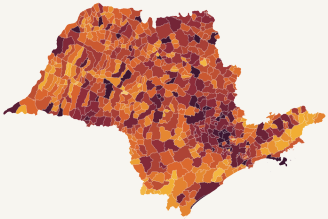
Grouping by	R^2 income/person
Stage C vectors	0.05–0.14
Land cover + lights	0.17–0.23
Raw satellite vectors	0.23–0.29

Why: Stage C predicts total GDP and subtracts population only at the end, so its vectors encode size. They are also unstable across trained models (adjusted Rand 0.45 vs 0.90 for raw vectors).

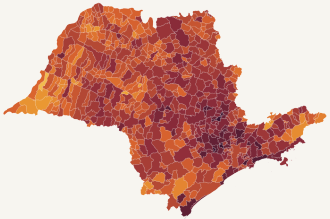
Trained in Minas Gerais, it still works in São Paulo

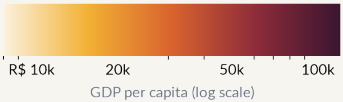
NEW SINCE MID-TERM

OFFICIAL, IBGE 2022

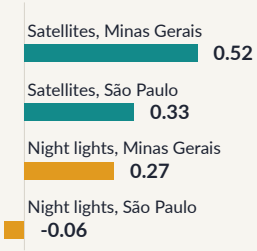


PREDICTED, NEVER TRAINED ON SP





R^2 AGAINST OFFICIAL GDP



645 municipalities, all models and scalers from Minas Gerais, nothing refitted. Rank agreement **0.61** (night lights 0.33); 66% within a factor 1.5. Predicted activity matches SP's real night lights (R^2 0.82).

What did not work, and what it taught us

WE TRIED	WHAT HAPPENED	LESSON
● Night-light steering	Works at cell level (neighbour check) but adds no GDP information	Light is already in the embeddings; calibration mattered more
● Text alignment	Small gain, shrank in the rerun ($-0.025 \rightarrow -0.009$)	Frozen embeddings limit what text can add
● POIs and roads	Nothing on top of text	The fact sentences already carry them
● Town vectors as an "economic space"	Twins match size, not wealth; clusters explain 0.05–0.14	Use the model's estimates, not vector distances
● Seed spread as uncertainty	Does not track the error ($\rho = 0.04$)	Seeds of one model are not a confidence interval

All checks out-of-sample; details in the report and reports/tables.

Answers to the five questions, and the limits

RESEARCH QUESTIONS: MID-TERM → FINAL

RQ1 night-light steering	NO CLEAR EFFECT
RQ2 text (captions)	SMALL GAIN, NOT ROBUST
RQ3 POIs and roads	NOTHING ADDED
RQ4 hybrid	BEST, NOT SIGNIFICANT
RQ5 where it fails	MINES, STEEL, COMMUTER TOWNS
Beyond: a new state	SATELLITES TRAVEL, LIGHTS DO NOT

Changed since the mid-term: RQ2 (from “largest gain” to “not robust”), RQ5 and the transfer answered.

LIMITS

- GSED is frozen: text can reorganise, not add, image information.
- Caption audit labelled by an LLM, one labeller, no agreement score yet.
- Stage C was revised after the first test results (choices made on validation).
- Trained on one state, tested on one more; other countries and years untested.
- Satellites cannot see the value of mines or industry.

TAKE-AWAY

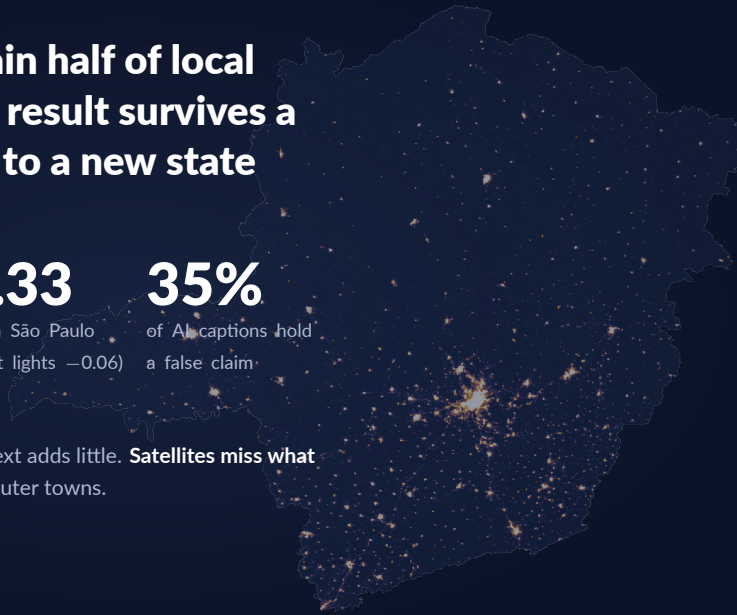
Free satellite data explain half of local income differences, the result survives a full rerun, and it travels to a new state where night lights fail.

0.52
R^2 in Minas Gerais, unseen regions

0.27
R^2, night lights alone

0.33
R^2 in São Paulo (night lights −0.06)

35%
of AI captions hold a false claim



Steering, POIs and roads add nothing clear; text adds little. **Satellites miss what they cannot see:** mines, steel mills and commuter towns.

Questions?

github.com/chadsaras/GIS_Project

Backup: two recent ideas, never tested together

UrbanCLIP Yan et al., WWW 2024

A language model writes a **caption for every satellite tile**; image and text are aligned.

– GDP gain only +0.5% R^2; random split lets neighbours leak into the test

Satellite embeddings Yue, Zhao & Hu, 2026

Google's **64-number embeddings**, **steered by night lights** towards economic meaning.

– countries only; no text, no POI or roads

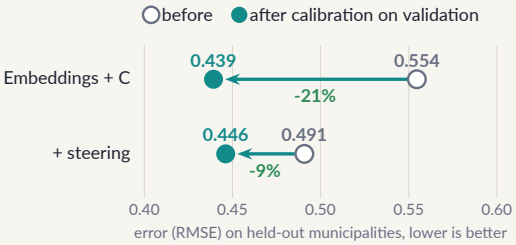
THIS PROJECT

Both, in one pipeline

● embeddings ● text
● night lights ● POI, roads

853 municipalities · spatial test ablation of every ingredient

Backup: calibration mattered more than night-light steering



1. **Predictions were over-spread.** A straight-line correction fitted on validation municipalities cut error by **−21%**.
2. **Honest testing changes conclusions.** Steering looked helpful before the fix; after it, the gain vanished.
3. **Disclosed:** the fix was chosen on validation data, but after the first test results were seen.

Backup: what predicts a large error?

NEW SINCE MID-TERM

Municipality trait	Hybrid (V10) coef. (SE)	p	Night lights (V1) coef. (SE)	p
official log GDP per capita	0.081 (0.035)	0.02	0.190 (0.037)	<0.001
built-up share	0.836 (0.473)	0.08	0.791 (0.250)	0.002
services share of the economy	-0.285 (0.132)	0.03	-0.332 (0.123)	0.007
industry share of the economy	0.185 (0.121)	0.13	-0.107 (0.102)	0.29
flagged (mining, hydro, top 1%)	0.165 (0.090)	0.07	0.272 (0.079)	<0.001
OSM completeness	-0.018 (0.082)	0.82	-0.099 (0.054)	0.07

OLS of absolute error, 853 municipalities, standard errors clustered by immediate region. Sector shares from 2021. IBGE publishes no data-quality score per municipality, so income and economic structure are used instead.